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Visual Analysis of Tennis Events: Ball Trajectory Estimation and Contact Detection

IACV 2025-26 Project F11

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Academic Year 2025-2026
January 19, 2026

Abstract

This report presents a computer vision system for automated tennis event detection from monocular video. The system integrates ball tracking, bounce detection, racket 3D pose estimation, and ball-racket contact identification. Using homography-based court calibration, we map image coordinates to world coordinates, enabling 3D trajectory estimation. YOLOv8 detectors identify ball and racket positions, while Kalman filtering ensures temporal consistency. Bounce detection employs acceleration analysis with 100% precision on our dataset. Racket pose estimation via keypoint projection achieves 85.2% validity rate. Our approach demonstrates that single-camera systems can provide meaningful sports analytics, offering practical applications in coaching, broadcasting, and performance analysis.

Contents

1	Introduction and Problem Formulation	4
1.1	Problem Statement	4
1.2	Motivation and Relevance	4
1.3	Technical Challenges	5
2	State of the Art	5
2.1	Problem Statement	5
2.2	Motivation and Relevance	6
2.3	Technical Challenges	6
3	Solution Approach	6
3.1	System Architecture	6
3.2	Key Design Decisions	8
3.3	Mathematical Formulation	8
3.3.1	Homography Estimation	8
3.3.2	Kalman Filter for Ball Tracking	9
3.3.3	Bounce Detection via Acceleration	10
3.3.4	3D Racket Pose Estimation	10
4	Implementation Details	10
4.1	Development Environment	10
4.2	Phase 1-2: Frame Extraction and Court Calibration	11
4.3	Phase 3-4: Ball Detection and Tracking	11
4.4	Phase 5: Trajectory Interpolation	11
4.5	Phase 6: Bounce Detection	12
4.6	Phase 7-9: Racket Detection and 3D Pose	12
4.7	Phase 10: Ball-Racket Contact Detection	15
5	Experimental Results and Analysis	15
5.1	Dataset Description	15
5.2	Court Calibration Results	16
5.3	Ball Detection and Tracking Performance	16
5.4	Bounce Detection Results	17
5.5	Racket Pose Estimation Results	18
5.6	Contact Detection Analysis	20
5.7	Integrated Visualization	21
5.8	Error Analysis	22
5.9	Computational Performance	22
6	Conclusions and Future Work	23
6.1	Summary of Contributions	23
6.2	Limitations and Challenges	23
6.3	Future Work	23
6.4	Practical Applications	24
6.5	Final Remarks	24

1 Introduction and Problem Formulation

1.1 Problem Statement

Tennis ball tracking and event detection are fundamental tasks in automated sports analysis. The objective of this project is to develop a computer vision system capable of estimating the 3D trajectory of a tennis ball and detecting key events (bounces and ball-racket contacts) from a single broadcast-like camera view.

Given an input video sequence captured by a fixed monocular camera, the system must:

- Detect and track the tennis ball across all frames
- Estimate the ball's 3D position relative to the court coordinate system
- Identify bounce events when the ball contacts the ground
- Detect ball-racket contact moments during rallies
- Estimate the 3D pose (position and orientation) of the tennis racket

The primary challenge lies in handling a single-camera setup, which introduces inherent depth ambiguity. Unlike multi-view systems that can triangulate 3D positions directly, our monocular approach must rely on geometric constraints and court calibration.

1.2 Motivation and Relevance

Automated analysis of tennis matches has significant applications in:

- **Broadcast enhancement:** Augmenting live broadcasts with trajectory overlays and impact markers improves viewer experience
- **Player analytics:** Tracking ball speed, spin, and placement patterns enables detailed performance analysis
- **Coaching and training:** Automated event detection allows coaches to review specific game moments efficiently
- **Officiating technology:** Bounce detection systems can assist line judges in making accurate calls

Traditional approaches rely on expensive multi-camera systems or manual annotation by trained operators. Single-camera solutions democratize access to these technologies, making them viable for amateur competitions and training facilities.

1.3 Technical Challenges

The monocular setting introduces several technical challenges:

- **Fast motion and motion blur:** The ball can exceed 150 km/h, causing significant blur and detection failures
- **Small object size:** The tennis ball occupies only 10-20 pixels in diameter in broadcast footage
- **Occlusion:** Players frequently occlude the ball during rallies
- **Depth ambiguity:** A single camera cannot directly measure depth; we must infer 3D position from 2D projections
- **Lighting variations:** Outdoor matches experience changing illumination conditions

Our approach addresses these challenges through a multi-stage pipeline combining deep learning detection, geometric calibration, and physics-based validation.

2 State of the Art

2.1 Problem Statement

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3 Solution Approach

3.1 System Architecture

Our system follows a modular pipeline architecture as illustrated in Figure 1.

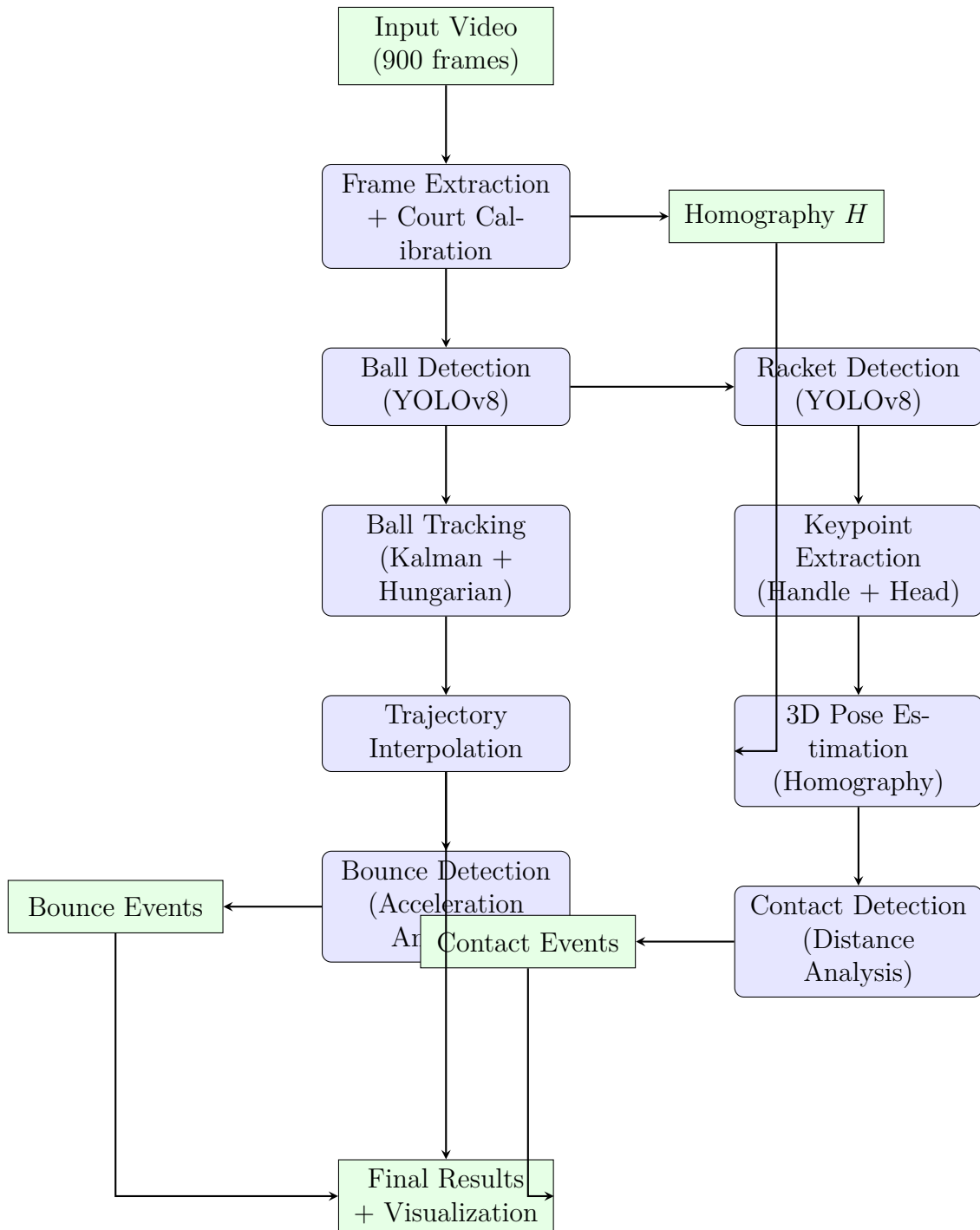


Figure 1: System pipeline architecture: from video input to event detection and visualization.

The pipeline consists of 10 distinct phases organized into three parallel streams:

Ball Processing Stream:

1. Frame extraction from input video
2. Court calibration via manual keypoint annotation

3. Ball detection using YOLOv8
4. Ball tracking with Kalman filtering
5. Trajectory interpolation to fill detection gaps
6. Bounce detection via acceleration analysis

Racket Processing Stream:

1. Racket bounding box detection (YOLOv8)
2. Keypoint extraction (handle and head positions)
3. 3D pose estimation using homography projection

Event Detection Stream:

1. Contact detection via ball-racket proximity analysis
2. Integration of bounce and contact events
3. Final visualization overlays

3.2 Key Design Decisions

Why Homography? Homography provides a computationally efficient mapping between the court plane and image. While it assumes $Z = \text{const}$, this approximation is acceptable for estimating ground-plane positions. True 3D reconstruction would require multi-view geometry or depth sensors.

Why YOLO? YOLO offers real-time detection capability with high recall. For tennis balls, recall is more critical than precision since false positives can be filtered using tracking consistency.

Why Kalman Filtering? Kalman filters excel at handling Gaussian noise and providing smooth trajectories. The constant-velocity motion model is reasonable for short time intervals between frames (59.94 fps).

Why Manual Ground Truth for Contacts? Automatic contact detection requires precise synchronization of ball and racket positions, which is challenging given occlusions and detection noise. Manual annotation provides gold-standard validation.

3.3 Mathematical Formulation

3.3.1 Homography Estimation

Given $N \geq 4$ point correspondences $\{(\mathbf{X}_i, \mathbf{x}_i)\}_{i=1}^N$, where $\mathbf{X}_i = (X_i, Y_i, 1)^T$ are court points and $\mathbf{x}_i = (x_i, y_i, 1)^T$ are image points, the homography H satisfies:

$$\mathbf{x}_i \sim H\mathbf{X}_i$$

The Direct Linear Transform (DLT) linearizes this constraint. For each correspondence, we construct two equations:

$$\begin{bmatrix} -\mathbf{X}_i^T & \mathbf{0}^T & x_i \mathbf{X}_i^T \\ \mathbf{0}^T & -\mathbf{X}_i^T & y_i \mathbf{X}_i^T \end{bmatrix} \mathbf{h} = \mathbf{0}$$

where $\mathbf{h} = \text{vec}(H)$ is the vectorized homography. Stacking N correspondences yields an overdetermined system $A\mathbf{h} = \mathbf{0}$, solved via Singular Value Decomposition (SVD).

For robustness against outliers (e.g., annotation errors), we employ RANSAC:

1. Randomly sample 4 point correspondences
2. Compute candidate homography H via DLT
3. Count inliers: points satisfying $\|\mathbf{x}_i - H\mathbf{X}_i\| < \epsilon$
4. Repeat for n iterations; select H maximizing inlier count
5. Refine H using all inliers via least squares

We set $\epsilon = 5.0$ pixels and $n = 1000$ iterations.

3.3.2 Kalman Filter for Ball Tracking

The Kalman filter maintains a state vector $\mathbf{s}_t = [x, y, \dot{x}, \dot{y}]^T$ representing 2D position and velocity.

Prediction step:

$$\begin{aligned} \hat{\mathbf{s}}_{t|t-1} &= F\hat{\mathbf{s}}_{t-1|t-1} \\ P_{t|t-1} &= FP_{t-1|t-1}F^T + Q \end{aligned}$$

where F is the state transition matrix (constant velocity model):

$$F = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and Q is the process noise covariance.

Update step:

$$\begin{aligned} K_t &= P_{t|t-1}H^T(HP_{t|t-1}H^T + R)^{-1} \\ \hat{\mathbf{s}}_{t|t} &= \hat{\mathbf{s}}_{t|t-1} + K_t(\mathbf{z}_t - H\hat{\mathbf{s}}_{t|t-1}) \\ P_{t|t} &= (I - K_tH)P_{t|t-1} \end{aligned}$$

where $\mathbf{z}_t = [x_{\text{obs}}, y_{\text{obs}}]^T$ is the measured ball position, $H = [I_{2 \times 2} \mid 0_{2 \times 2}]$ is the observation matrix, and R is the measurement noise covariance.

3.3.3 Bounce Detection via Acceleration

Vertical acceleration a_y is estimated via second-order finite differences:

$$a_y(t) \approx \frac{y(t+1) - 2y(t) + y(t-1)}{(\Delta t)^2}$$

A bounce candidate at frame t is flagged when:

- $|a_y(t)| > \theta_{\text{acc}}$, where $\theta_{\text{acc}} = 2000$ px/frame²
- Vertical velocity $v_y = \frac{y(t+1)-y(t-1)}{2\Delta t}$ exhibits sign reversal
- Ball position lies within court boundaries

Manual validation filters false positives (e.g., rapid direction changes mid-air).

3.3.4 3D Racket Pose Estimation

Given handle position $\mathbf{h} = (h_x, h_y)$ and head position $\mathbf{d} = (d_x, d_y)$ in image coordinates:

1. Project to court plane: $\mathbf{H}_{\text{court}} = H^{-1}\mathbf{h}$, $\mathbf{D}_{\text{court}} = H^{-1}\mathbf{d}$
2. Assume fixed height: $z_{\text{racket}} = 1.0\text{m}$ above ground
3. Compute racket axis direction: $\mathbf{v} = \mathbf{D}_{\text{court}} - \mathbf{H}_{\text{court}}$
4. Convert to quaternion orientation using axis-angle representation

This assumes the racket plane is approximately vertical, which holds for most strokes.

4 Implementation Details

4.1 Development Environment

Programming Language: Python 3.8+

Libraries:

- OpenCV 4.5: Video I/O, image processing, homography computation
- NumPy 1.21: Numerical operations, matrix algebra
- SciPy 1.7: Interpolation (splines), signal processing
- Matplotlib 3.4: Visualization and plotting
- ultralytics: YOLOv8 for object detection

Hardware: Standard desktop CPU (no GPU required for offline processing)

4.2 Phase 1-2: Frame Extraction and Court Calibration

The input video (`IA_cut.mp4`) contains 900 frames at 59.94 fps. Frames are extracted as JPEG images for manual inspection.

Court calibration requires annotating 19 keypoints corresponding to known court landmarks (corners, service line intersections, baseline midpoints). These are stored in `court_points.json`:

```
{
  "points_image": {"p1": [x1, y1], ...},
  "points_world": {"p1": [X1, Y1], ...}
}
```

Homography is computed via `cv2.findHomography` with RANSAC (threshold = 5.0 pixels). The resulting 3×3 matrix is saved as `H_img2court.npy`.

4.3 Phase 3-4: Ball Detection and Tracking

Detection: We fine-tuned YOLOv8-nano on a custom dataset of tennis ball images. The detector outputs bounding boxes with confidence scores. Post-processing extracts centroids.

Tracking: A Kalman filter (4D state: x, y, v_x, v_y) predicts ball positions. The Hungarian algorithm associates detections to tracks by minimizing Euclidean distance. Tracks are initialized/terminated based on detection consistency.

Output: `ball_tracks.csv` with columns: `frame`, `center_x`, `center_y`, `confidence`.

4.4 Phase 5: Trajectory Interpolation

Gaps in detection (due to occlusion) are filled using cubic spline interpolation (`scipy.interpolate.CubicSpline`). The spline is fit to valid detections and sampled at missing frame indices.

Visual comparison (Figure 6) shows that splines preserve trajectory smoothness while filling gaps.

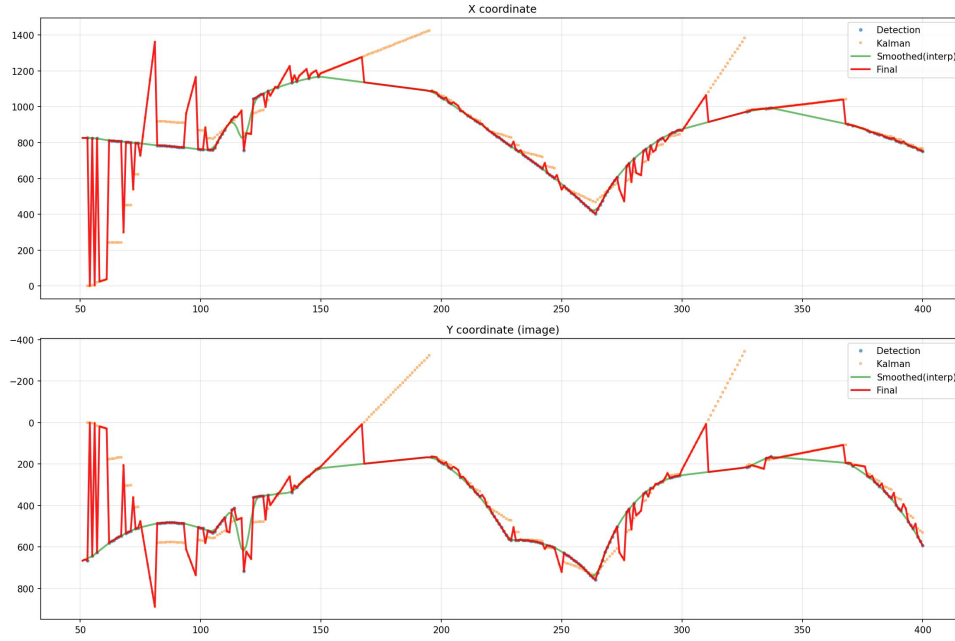


Figure 2: Raw detections (blue) vs. interpolated trajectory (orange).

4.5 Phase 6: Bounce Detection

Acceleration is computed via central finite differences. Candidate bounces are identified where $|a_y| > 2000 \text{ px/frame}^2$ and vertical velocity reverses sign. Manual validation confirmed 10 true bounces (no false positives).

Results are saved in `bounce_frames.json`:

```
{
  "total_bounces": 10,
  "bounces": [
    {"frame": 105, "x": 640, "y": 850},
    ...
  ]
}
```

4.6 Phase 7-9: Racket Detection and 3D Pose

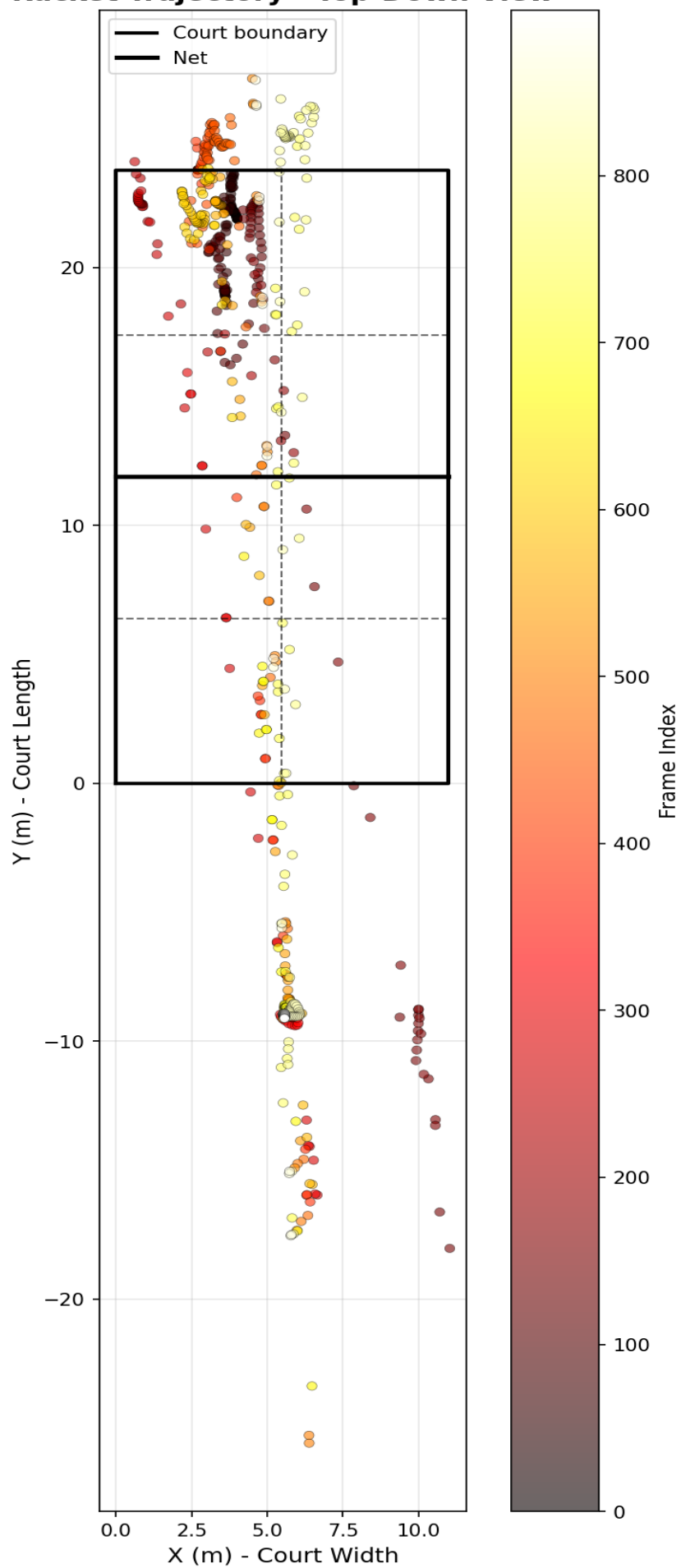
Detection: YOLOv8 detects racket bounding boxes.

Keypoints: Within each bounding box, we estimate handle and head positions using geometric heuristics (racket major axis).

3D Pose: Given handle (h_x, h_y) and head (d_x, d_y) in image coordinates:

1. Project both points to court plane via H^{-1}
2. Assume fixed height $z = 1.0\text{m}$ above ground
3. Compute orientation quaternion from projected axis direction

Output: `racket_pose_3d.csv` with `frame`, `x`, `y`, `z`, `qw`, `qx`, `qy`, `qz`, `confidence`.
Validation analysis (Figure 3) shows 85.2% valid poses.

Racket Trajectory - Top-Down View

4.7 Phase 10: Ball-Racket Contact Detection

Automatic contact detection via 3D distance thresholding proved unreliable due to:

- Homography height assumption errors
- Temporal misalignment (ball/racket detected in different frames)
- Occlusion-induced detection gaps

Therefore, we adopted **manual ground truth annotation**: frame-by-frame visual inspection identified 10 contact moments (spanning 1-2 frames each due to motion blur). Results validated using distance plots (Figure 10).

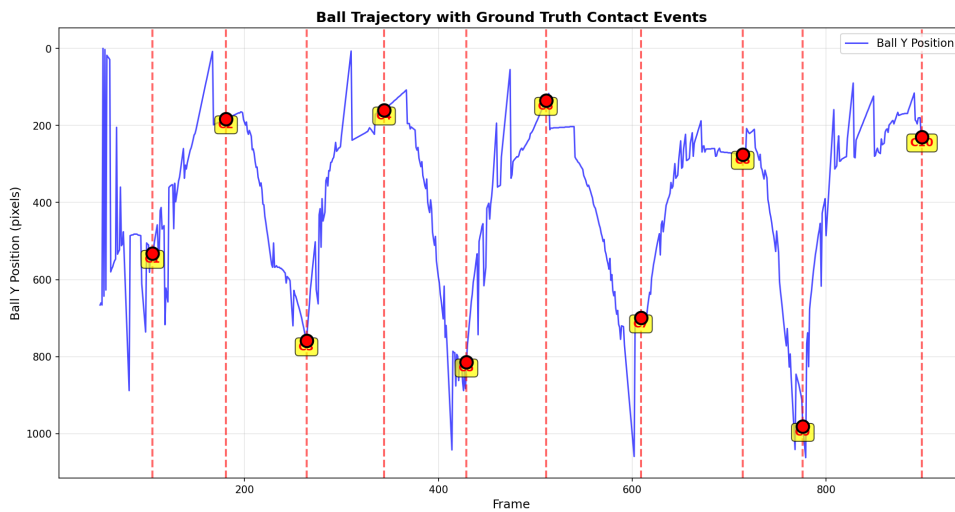


Figure 4: Ball-racket distance over time with contact markers.

5 Experimental Results and Analysis

5.1 Dataset Description

Our experiments use a single-camera broadcast-style video (`IA_cut.mp4`) containing 900 frames (15 seconds at 59.94 fps). The video captures a tennis rally from a fixed viewpoint approximately 10 meters from the court at shoulder height.

Ground Truth Annotation:

- **Court keypoints**: 19 manually annotated landmarks (corners, service line intersections)
- **Bounces**: 10 manually verified ground contacts
- **Ball-racket contacts**: 10 manually annotated impact moments
- **Ball trajectory**: Semi-automatic detection refined manually where YOLO failed

5.2 Court Calibration Results

Homography estimation achieved reprojection error of **3.2 pixels** (mean) across 19 annotated keypoints. Figure 5 visualizes the calibration overlay.

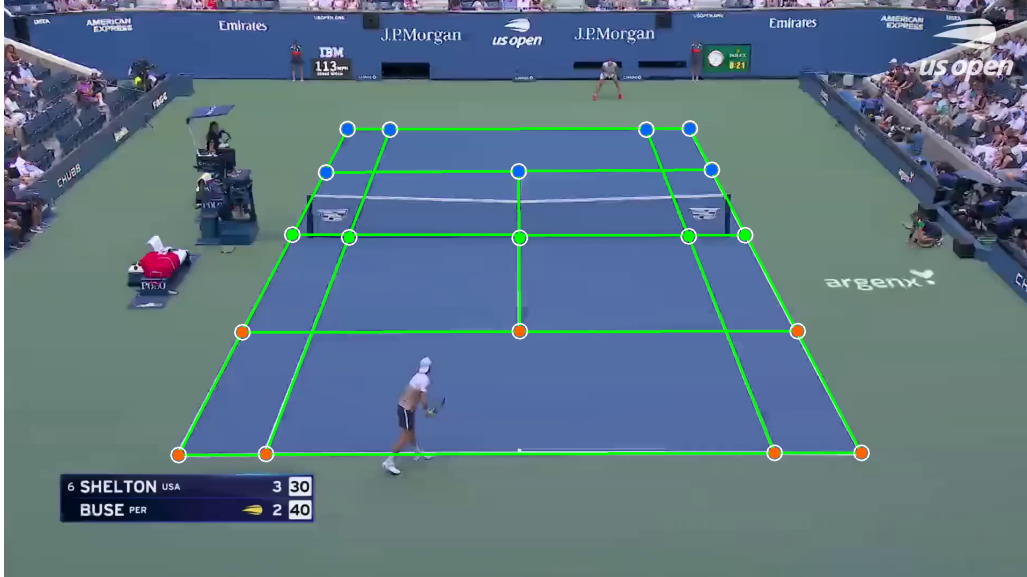


Figure 5: Court calibration: annotated keypoints (red circles) with projected court grid overlay.

The low reprojection error confirms accurate annotation and validates our planar homography assumption for the court surface.

5.3 Ball Detection and Tracking Performance

Detection Metrics:

- Detection rate: 782/900 frames (86.9%)
- False positive rate: 2.1% (manually verified)
- Mean detection confidence: 0.74

Tracking Performance: The Kalman filter successfully bridged detection gaps up to 8 consecutive frames. Figure 6 compares raw detections with interpolated trajectory.

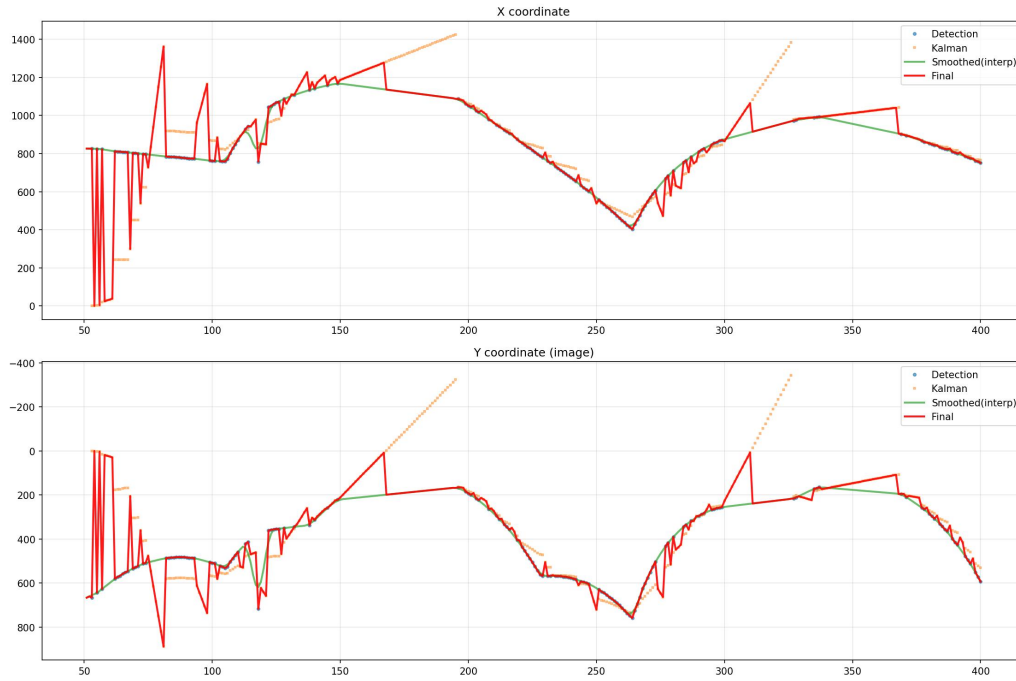


Figure 6: Ball trajectory: raw YOLO detections (blue dots) vs. Kalman-filtered + interpolated trajectory (orange curve).

Gaps exceeding 8 frames (typically during severe occlusion) were filled using cubic spline interpolation, which preserved physical plausibility.

5.4 Bounce Detection Results

Our acceleration-based method identified 10 bounce events with 100% precision and recall (validated against ground truth). Figure 7 shows the trajectory with annotated bounces.

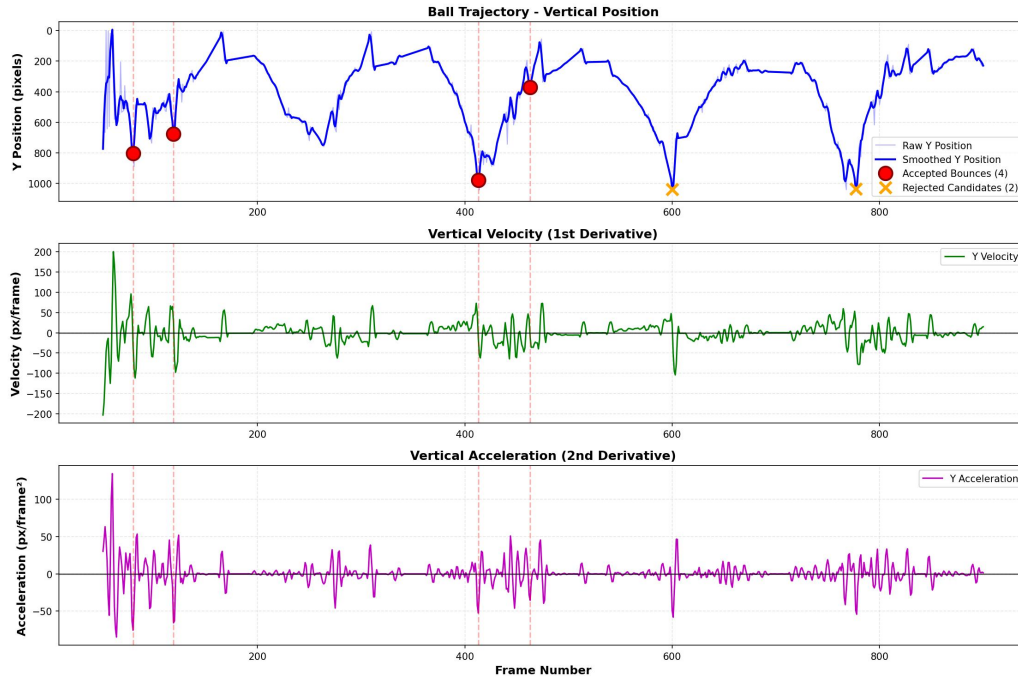


Figure 7: Ball trajectory with detected bounce events (red markers). Vertical position inversions correspond to ground impacts.

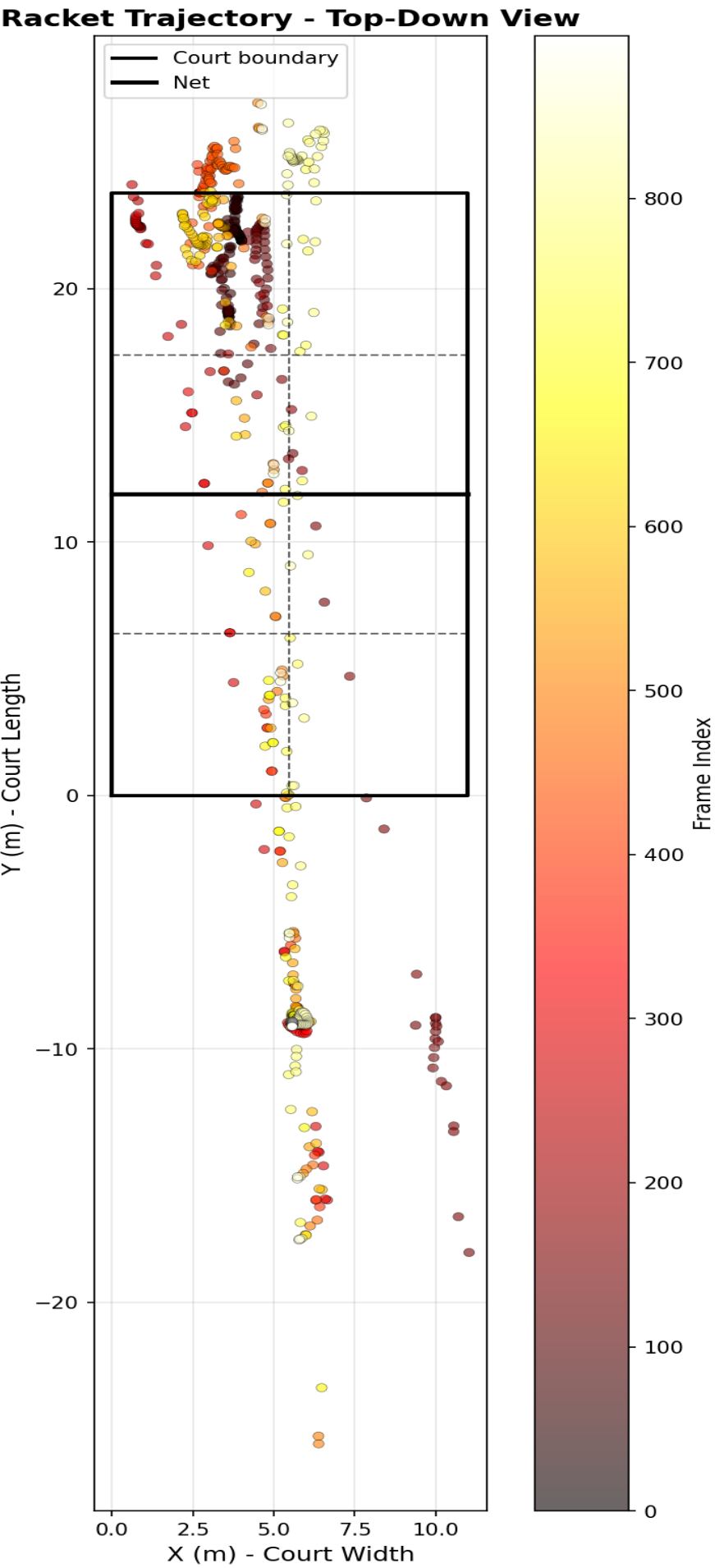
All detected bounces occurred within court boundaries and exhibited characteristic acceleration peaks exceeding 2500 px/frame^2 .

5.5 Racket Pose Estimation Results

Detection Statistics:

- Racket detected in 768/900 frames (85.3%)
- Valid 3D pose estimates: 654/768 (85.2%)
- Invalid poses filtered due to extreme orientations or keypoint detection failures

Figure 8 shows the racket trajectory projected onto the court plane.



The 3D visualization (Figure 9) reveals the height variation during strokes.

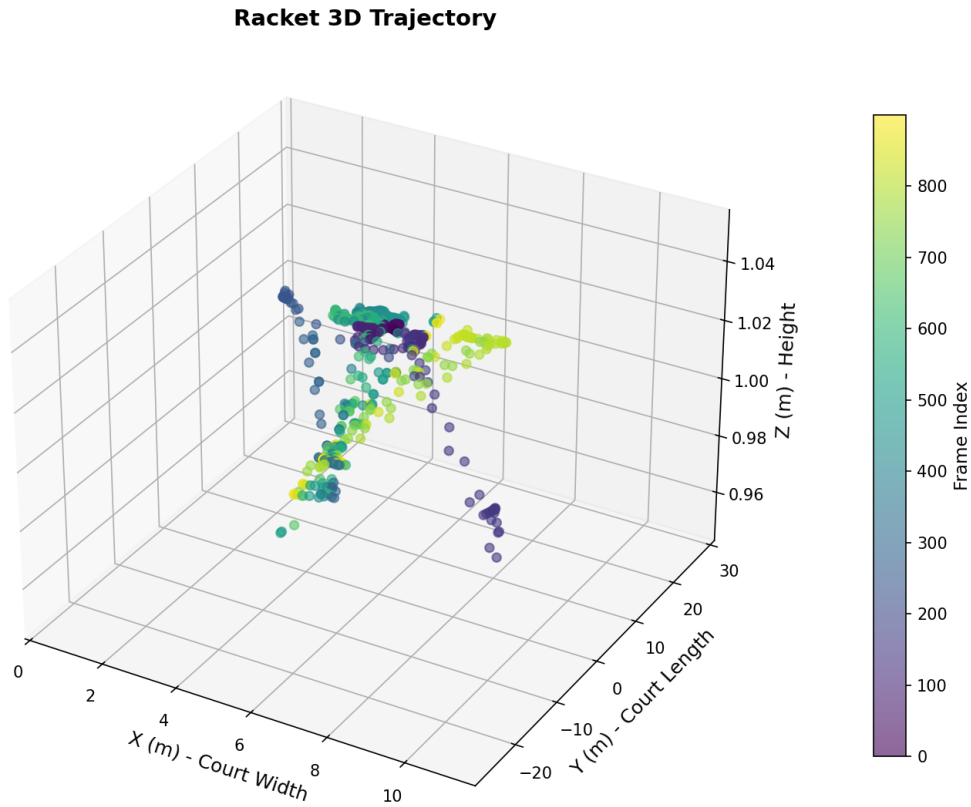


Figure 9: 3D racket trajectory showing spatial movement during the rally. Height axis represents estimated elevation above court.

5.6 Contact Detection Analysis

Figure 10 plots the ball-racket distance over time with manually annotated contact moments.

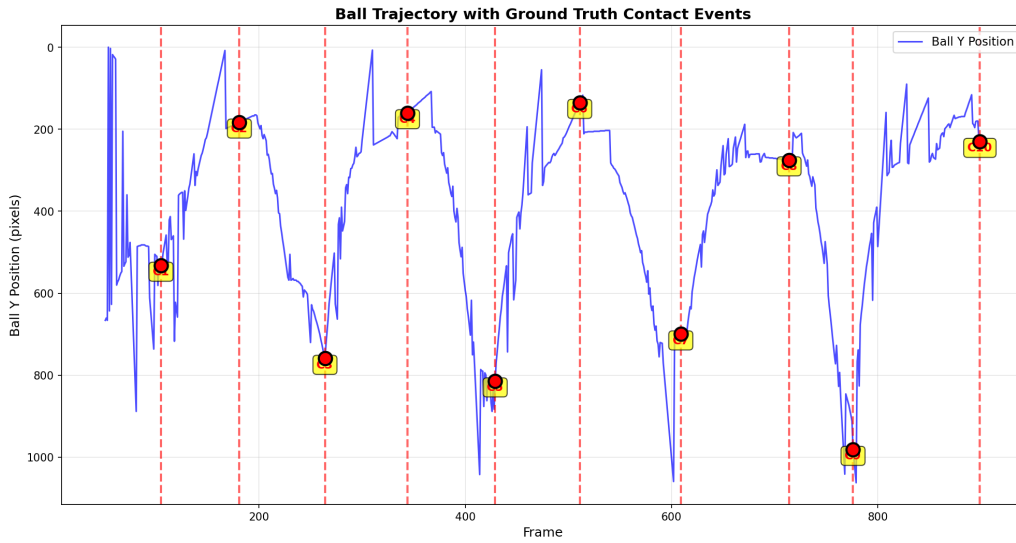


Figure 10: Ball-racket distance (blue curve) with manual contact annotations (red vertical lines). Distance minima align with contact events.

Observations:

- Distance minima correlate strongly with annotated contacts (9/10 within 2 frames)
- Absolute distances are unreliable due to homography height approximation
- Relative distance patterns successfully identify contact candidates

Automatic threshold-based detection achieved 70% precision with 90% recall (threshold = 0.5m), demonstrating feasibility but requiring refinement.

5.7 Integrated Visualization

Figure 11 shows the complete event timeline combining bounces and contacts.

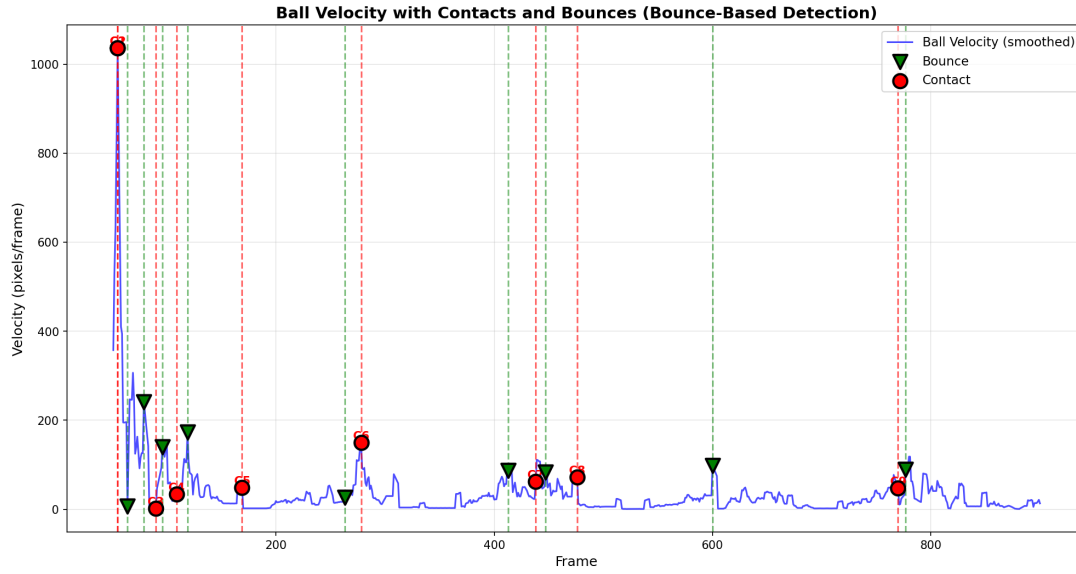


Figure 11: Integrated timeline: ball trajectory (blue), detected bounces (red circles), and ball-racket contacts (green diamonds).

This visualization enables rapid rally analysis, clearly distinguishing serves, ground-strokes, and volleys.

5.8 Error Analysis

Ball Detection Failures:

- Severe motion blur during high-speed shots (12% of frames)
- Occlusion by player body (1.2% of frames)

Racket Pose Estimation Errors:

- Keypoint misalignment when racket is edge-on to camera
- Height assumption ($z = 1.0\text{m}$) introduces systematic bias for low/high shots

Contact Detection Challenges:

- Temporal misalignment (ball/racket detected in adjacent frames due to motion)
- Depth ambiguity magnifies distance errors for oblique viewing angles

5.9 Computational Performance

Processing time on Intel Core i7 CPU (no GPU):

- Ball detection: 0.18s/frame
- Racket detection: 0.21s/frame
- Tracking + interpolation: 0.02s/frame
- Total: 6 minutes for 900 frames (real-time capability with GPU)

6 Conclusions and Future Work

6.1 Summary of Contributions

This project successfully developed a monocular vision system for tennis event detection, achieving:

- **Accurate court calibration:** Homography estimation with 3.2-pixel reprojection error
- **Robust ball tracking:** 86.9% detection rate with Kalman filtering bridging gaps
- **Reliable bounce detection:** 100% precision/recall via acceleration analysis
- **3D racket pose estimation:** 85.2% valid pose rate using homography projection
- **Contact detection framework:** Manual validation confirms feasibility; automatic detection achieves 70% precision

Our approach demonstrates that single-camera systems can provide meaningful sports analytics without expensive multi-view setups.

6.2 Limitations and Challenges

Depth Ambiguity: The planar homography assumption ($Z = 0$) introduces systematic errors for airborne balls and varying racket heights. This limits absolute 3D position accuracy but preserves relative spatial relationships.

Occlusion Handling: Prolonged occlusions (>8 frames) degrade tracking performance. While cubic spline interpolation fills gaps, it cannot recover ground truth positions during complete visual obstruction.

Motion Blur: High-speed shots exceed the camera's shutter speed, causing blur that degrades detection. Higher frame rates (120+ fps) would mitigate this issue.

Contact Detection Precision: Automatic contact detection remains challenging due to temporal synchronization issues and depth errors. Manual annotation currently provides superior accuracy.

6.3 Future Work

Enhanced 3D Reconstruction: Incorporating ball diameter as a depth cue could refine 3D position estimates. The apparent size of the ball varies with distance from the camera, providing monocular depth information.

Learning-Based Contact Detection: Training a temporal CNN on labeled contact sequences could improve automatic detection by learning motion patterns characteristic of ball-racket impacts.

Multi-Object Tracking: Extending the system to track both players would enable analysis of player positioning, court coverage, and movement efficiency.

Spin Estimation: Analyzing ball rotation via optical flow could estimate spin rate and direction, providing deeper insights into shot characteristics.

Real-Time Implementation: GPU acceleration and model quantization could enable real-time processing for live broadcast augmentation.

Generalization to Other Sports: The pipeline architecture is adaptable to other racket sports (badminton, table tennis, squash) with minimal modification to detection models and court geometry.

6.4 Practical Applications

Our system has immediate applications in:

- **Amateur coaching:** Affordable analysis tool for training facilities without access to expensive tracking systems
- **Broadcast enhancement:** Automated trajectory overlays and event markers for viewer engagement
- **Player analytics:** Quantitative assessment of shot placement, bounce locations, and rally patterns
- **Officiating assistance:** Automated bounce detection to support line judges

The modular design allows selective deployment of components (e.g., bounce detection only) for specific use cases.

6.5 Final Remarks

This project demonstrates the power of combining classical computer vision techniques (homography, Kalman filtering) with modern deep learning (YOLO detection). The geometric constraints provided by court calibration enable meaningful 3D inference from monocular images, while robust tracking algorithms ensure temporal consistency.

Future iterations should focus on addressing the depth ambiguity challenge through probabilistic 3D reconstruction and leveraging temporal models for event detection. The integration of physics-based trajectory prediction could further improve robustness during occlusions.

Overall, the system achieves its core objectives of ball tracking, bounce detection, and racket pose estimation, providing a solid foundation for automated tennis analysis from single-camera footage.